

Project description:

This project was aimed to implement a fully working and functional tic tac toe game. If you aren't familiar, tic tac toe is a classic 2-player strategy based on a 3x3 grid where the players take turns marking spaces with either an "X" or an "O". the winner is the player that can mark 3 adjacent spaces with either an "X" or an "O".

Approach and logic used:

I approached this project with 3 core functions: *print_board()*, *check_winner()*, and *is_draw()*. In my mind, I thought the best way to approach the project was to base my code around these 3 core functions and add a function that tracks the user's moves then have it run through these 3 functions to update the game and make sure it isn't supposed to end.

Data structures used:

The primary data structure used in this project is a 2D list to represent the game board. Each element in the list corresponds to a box on the board and stores either a blank space, "X", or "O". Alongside the board, variables are used to track the current player, player symbols, and the game state (over, not over)

Challenges faced:

One of the main challenges in this project was implementing a reliable method to check for winning conditions. Another challenge was handling user input. In my code I tried to handle with with a try and except to check for a ValueError but in the end I decided to a long if statement since I was having trouble using except.

Possible improvements

One major improvement would be adding an AI opponent, allowing a single player to compete against the computer, one easy way this could be implemented is just having the computer pick at random. The program could also be extended to support replaying each other multiple rounds without restarting the program.